Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$P = \left(\begin{array}{ccccc} P_{(+++)} & & & & \\ & P_{(+-+)} & & \text{\huge 0} & \\ & & P_{(+--)} & & \\ & \text{\huge 0} & & \ddots & \\ & & & & P_{(---)} \end{array} \right)$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\left(\begin{array}{ccc} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{array} \right)_q = \left(\begin{array}{ccc} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{array} \right)_q = (-)^{j_1+j_2+j_3} \left(\begin{array}{ccc} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{array} \right)_{q^{-1}}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\int_{I_2} |\hat{\psi}(r_\xi(\omega))|^2 \beta(\omega)\, d\omega \le \int_{I_2} C^2 (1+|r_\xi(\omega)|)^{-2r}\, d\omega \\ &\quad \le \int_{I_2} C^2 (1+|r_\xi(\omega_\xi^*)|)^{-2r}\, d\omega \\ &\quad = \frac{\alpha}{1-\alpha} C^2 \xi^\alpha (1+|r_\xi(\omega_\xi^*)|)^{-2r}. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &\omega^2(x) = -\mathrm{Tr}\Big[\Big(W'(\hat{\Phi}) + [\hat{X}, \hat{Y}]\Big) \tfrac{1}{x-\hat{\Phi}}\Big] \ , \\ &\omega\Big(x + i\tfrac{m}{2}\Big)\omega\Big(x - i\tfrac{m}{2}\Big) = \mathrm{Tr}\left(\hat{Y}\hat{X} \tfrac{1}{x-\hat{\Phi}-i\frac{m}{2}} - \hat{X}\hat{Y} \tfrac{1}{x-\hat{\Phi}+i\frac{m}{2}} \right) \ . \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{rclcrcl} \psi'_- & = & e^{i\theta(x)}\psi_- & \qquad & \psi'_+ & = & \psi_+ \\ \psi'^{\dagger}_- & = & \psi^{\dagger}_- e^{-i\theta(x)} & & \psi'^{\dagger}_+ & = & \psi^{\dagger}_+ \\ \lambda_- & = & \lambda_-(\theta) & & & & , \end{array}$$